Effective methods and practices for documenting data

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PRUK Workshop ‘Enhancing longitudinal data sharing: Data management strategies and best practices’

25/11/2024

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Introduction

Good data documentation is essential to ensuring data can be used responsibly and correctly in future research. New users wanting to use data or researchers wishing to return to their data later, all need sufficient contextual and explanatory information to make sense of the data. Further, documenting data for longitudinal studies also ensures that data are comparable between waves.

Good data documentation makes data clear to understand and easy to use. At the heart of making data understandable and user-friendly is the foundation of solid metadata practices. Metadata helps users find, understand, and use data. It plays a critical role in data management, enabling better organisation, discoverability, and interoperability of data.

While data documentation can feel like an unnecessary task that takes up time and resources, if it is planned from the start, it will become part of the data lifecycle and easier to manage. It should be an ongoing process rather than an add-on task at the end.

We will look at:

- **Good documentation will make your data FAIR**
- **Different types of data documentation** (Data-level and study- or project-level)
- **Quantitative Data-level documentation** (Variable names, variable labels, value labels, standard classifications and coding schemes, missing values, derived variables)
- **Qualitative Data-level documentation** (Embedded metadata, data list)
- **Study-level documentation** (Best practice, UKDS example)
- **Metadata** (What is metadata, creating metadata during the research data lifecycle, structured metadata, metadata standards, example metadata schemes, Data Documentation Initiative (DDI), DDI and archive catalogue records including metadata profiles).

It is worth mentioning these topics are applicable to all data types, not just longitudinal data. However, they are particularly important to longitudinal data to ensure good practice is replicated with each new wave of data.

1. Good documentation will make your data FAIR

We have already introduced the FAIR principles but it is important to acknowledge that good data documentation is crucial to making your data FAIR. Using metadata, specifically a structured metadata scheme such as DDI, helps your data to comply with the FAIR principles. If we look at the FAIR principles, you will see how many times **metadata** is mentioned. This highlights how important metadata is to making data Findable, Accessible, Interoperable, and Reusable.

Findable:

- F1. **(Meta)data** are assigned a globally unique and persistent identifier
- F2. Data are described with rich **metadata**
- F3. **Metadata** clearly and explicitly include the identifier of the data they describe
- F4. **(Meta)data** are registered or indexed in a searchable resource

Accessible:

- A1. **(Meta)data** are retrievable by their identifier using a standardised communications protocol
    - A1.1 The protocol is open, free, and universally implementable
    - A1.2 The protocol allows for an authentication and authorisation procedure, where necessary
- A2. **Metadata** are accessible, even when the data are no longer available

Interoperable:

- I1. **(Meta)data** use a formal, accessible, shared, and broadly applicable language for knowledge representation.
- I2. **(Meta)data** use vocabularies that follow FAIR principles
- I3. **(Meta)data** include qualified references to other **(meta)data**

Reusable:

- R1. **(Meta)data** are richly described with a plurality of accurate and relevant attributes
- R1.1. **(Meta)data** are released with a clear and accessible data usage license
- R1.2. **(Meta)data** are associated with detailed provenance
- R1.3. **(Meta)data** meet domain-relevant community standards

2. Different types of data documentation

Data documentation can be divided into two levels:

Data-level documentation provides information on the individual data objects, such as a variable in a data file or an interview transcript. It can be embedded in the data file, such as variable or value labels in a data file, or participant information added in the header or an interview transcript.

Study-level documentation (sometimes referred to as project-level) provides high-level information on the research context and design, the data collection methods used, any data preparations and manipulations, plus summaries of findings based on the data.

Both types of data documentation are equally important to ensuring that research data can be shared and reused by a wide range of users for a variety of purposes.

We are going to look at both data-level documentation and study-level documentation in more detail and by data type, starting with quantitative data-level documentation.

Activity: Data without Context (Quantitative data)

Please have a look at this extract from a data file and take a few minutes to try and make sense of the data. What information is missing?

Exercise – Data without context (quant.)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1	ID	weight	bkweight	region	ethnicity	religion	hlqual	income	hhchildre	hhcomp	hhgens	cob	yar	lenres	belong	cbelong	ethimp	relimp	demethrel	natid	feelbri
2	1	4350.442	1.577572	7	14	2	2	13	0	5	2	1	1	6	2	2	3	3	5	1	1
3	2	6327.599	1.4787	10	14	2	2	4	0	7	1	1	1	5	2	1	3	1	4	3	3
4	3	2695.431	0.69719	9	14	2	2	17	0	3	1	1	1	8	1	3	1	1	4	2	2
5	4	10132.55	2.622927	3	14	1	5	4	0	5	2	1	1	5	3	3	4	6	5	1	3
6	5	744.6434	0.184362	9	18	9	6	16	1	6	3	1	1	4	3	5	1	6	4	1	3
7	6	1290.339	0.188772	8	5	1	1	15	0	3	1	1	1	1	1	3	2	6	5	1	1
8	7	823.8446	0.53351	9	6	2	6	12	1	4	2	41	6	3	2	5	3	3	5	6	2
9	8	1785.362	0.179279	2	9	1	4	2	0	7	1	1	1	2	4	5	3	6	5	1	1
10	9	4982.731	1.172688	2	14	1	4	4	0	1	1	1	1	6	4	4	3	6	5	6	2
11	10	778.6472	0.035212	9	1	6	4	7	0	5	1	1	1	4	2	3	2	2	4	1	2
12	11	1522.9	0.53251	8	14	1	1	14	0	3	1	1	2	5	5	3	5	6	5	1	1
13	12	89.98021	0.001393	2	5	2	1	5	1	7	1	1	1	1	2	3	2	2	5	1	1
14	13	286.4336	0.082532	3	18	2	2	12	0	1	1	1	1	6	1	3	4	4	5	6	5
15	14	11561.13	2.041249	3	14	1	4	7	0	3	2	1	1	5	1	1	4	6	5	1	1
16	15	2562.42	1.122225	7	14	2	2	5	0	3	1	1	1	2	2	2	2	1	4	2	2
17	16	807.3569	0.900323	8	18	1	1	13	1	4	2	36	6	1	1	3	3	6	5	6	5
18	17	7279.377	1.425031	5	14	2	2	8	2	2	2	1	1	4	2	2	2	5	6	1	2
19	18	2448.971	0.282259	9	13	2	3	5	0	3	1	33	7	1	4	5	2	1	4	6	2
20	19	5528.996	0.830848	7	14	1	2	8	0	1	1	3	1	4	4	2	1	6	4	3	2
21	20	1394.561	0.236408	9	3	4	1	11	2	6	2	17	6	2	1	2	1	1	2	1	3
22	21	2912.568	0.295174	9	20	1	2	12	1	6	3	1	1	6	2	3	2	6	5	1	2
23	22	22370.37	7.305734	6	14	2	4	13	0	3	2	1	1	8	2	3	3	3	2	1	2
24	23	228.0955	0.019548	11	12	2	1	7	0	3	1	50	7	1	2	2	3	3	5	6	5
25	24	951.1559	0.097839	11	18	9	3	17	0	1	5	51	1	1	4	4	4	6	4	1	1
26	25	524.2137	0.043126	2	3	6	1	5	3	2	1	1	1	2	2	3	2	1	2	1	2
27	26	66.7248	0.00131	9	2	1	1	14	0	1	1	13	7	2	2	4	2	6	5	6	4
28	27	2770.995	0.66842	4	14	2	2	8	0	3	1	1	1	8	1	1	2	3	4	2	2
29	28	76.82072	0.018688	9	21	5	1	17	0	1	1	1	1	6	2	2	1	1	2	1	1
30	29	352.5024	0.196928	8	1	1	4	16	0	5	1	1	1	1	4	6	1	6	4	2	4
31	30	158.0146	0.326188	3	3	7	2	17	0	3	1	17	3	8	1	3	1	1	3	1	1
32	31	2403.691	1.375405	9	18	2	2	12	1	4	2	1	1	6	2	2	2	3	5	1	2
33	32	363.9591	0.019879	9	2	1	1	11	0	3	1	1	1	3	2	3	1	6	4	2	1
34	33	4646.54	2.436886	5	14	1	2	9	0	3	1	1	1	3	2	1	2	6	2	1	2
35	34	6072.959	1.961362	5	14	1	1	14	0	5	2	1	1	6	1	3	2	6	5	1	1

Source: University of Manchester, Cathie Marsh Institute for Social Research (CMIST), UK Data Service. (2024). Evidence for Equality National Survey: a Survey of Ethnic Minorities During the COVID-19 Pandemic, 2021: Teaching Dataset. [data collection]. Finney, N., University of St Andrews, Department of Geography and Sustainable Development. SN: 9249, DOI: 10.5255/UKDA-SN-9249-1

Answers should include:
Variable labels
Value labels
Weighting information
Survey questions / questionnaire
Missing value information
Dates of fieldwork
Method of data collection
Sampling procedures
Population
Derived variables
User guide
Technical report
Data dictionary
Codebook.
(All should be forms of documentation - data-level, metadata or study/project-level.)

It becomes clear in the exercise, that accompanying documentation materials are essential. Two of the recommended documentation materials are a Data Dictionary, and a User Guide (snippets of those you can see below).

Exercise – Data without context (quant.)

Data dictionary extract

Name	Description	Category
region	Region of Great Britain	1 North East 2 North West 3 Yorkshire and Humber 4 West Midlands 5 East Midlands 6 East of England 7 South West 8 South East 9 London 10 Wales 11 Scotland
ethnicity	Ethnic group	1 Asian: Bangladeshi 2 Asian: Chinese 3 Asian: Indian 4 Asian: Pakistani 5 Asian: Any other Asian background 6 Black: African 7 Black: Caribbean 8 Black: Any other black background 9 Mixed: White and Asian 10 Mixed: White and black African 11 Mixed: White and black Caribbean 12 Mixed: Any other mixed group 13 White: Eastern European 14 White: English / Welsh / Scottish 15 White: Gypsy / Traveller 16 White: Irish 17 White: Roma 18 White: Any other white group 19 Other: Arab 20 Other: Any other ethnic background 21 Jewish
religion	Religion	1 No religion 2 Christian 3 Buddhist 4 Hindu 5 Jewish 6 Muslim 7 Sikh 8 Any other religion 9 Prefer not to say 10 Agnostic

User Guide extract

Introduction

EVENS is a national survey that documents the experiences and attitudes of ethnic and religious minorities in Britain. EVENS was developed by the Centre on the Dynamics of Ethnicity (CoDE) in response to the disproportionate impacts of COVID-19 and is the largest and most comprehensive survey of the lives of ethnic and religious minorities in Britain for more than 25 years. EVENS used pioneering, robust survey methods to collect data in 2021 from 14,200 participants of whom 9,700 identify as from an ethnic or religious minority.

The extract from the Data Dictionary illustrates what it provides: a listing of all variable names and labels, and their category values and labels. By just providing the variable and value labels we are now able to start to understand the data. Without this information, the data could not be used in a meaningful way.

The extract from the Introduction of the User Guide gives us a bit of background information about the data source. We still need more information, but even this small snippet begins to put the data into context.

Summary
This exercise highlights how documentation is essential to ensuring data can be understood and hence, reused. Users need sufficient contextual and explanatory information to make sense of the data.

3. Quantitative Data-level documentation

Quantitative data-level documentation generally relates to the variables in the data file. It can be embedded in the data, as many data software packages have facilities for data annotation and description as variable attributes (labels, codes, data type, missing values), table relationships, etc. Alternatively, information about data items can be recorded in a structured document such as a codebook. It is good practice to create this in addition to the embedded documentation as this will useful for transfer between different file formats.

We are going to look at best practice for some of the common elements of quantitative data-level documentation which you are likely to encounter.

3.1 Variable names

Variable names are important because they are used by the analysis software so they need to meet certain standards. Your choice of software might have specific standards, but some general rules are:

Variable names should

- not contain spaces
- start with a letter and not a number
- not contain any question or exclamation marks or a special character such as #, &, $, @ as these characters can interfere with specific software functions
- not be too long - eight characters is considered the ideal maximum length
- not use diacritics (marks above or below a letter) or national specific characters.

There are three simple approaches to naming variables:

- Using numeric codes that reflect the variable's position in a system (e.g. V001, V002, V003...).
- Using codes that refer to the research instrument (e.g. question number in a questionnaire: Q1, Q2, Q3a, Q3b...).
- Using mnemonic names that refer to the content of variables (e.g. AGE for respondent's age, ETHNIC for respondent's ethnicity etc.).

CESSDA Training Team (2017 - 2022). CESSDA Data Management Expert Guide. Bergen, Norway: CESSDA ERIC.

We find the mnemonic approach is commonly used. Users find meaningful and self-explanatory variable names easier to use. Sometimes a mixture of the above approaches are used, so survey question numbers might be used for the main content of the data, but the demographic variables are named using the mnemonic method.

3.1.1 Variable names and longitudinal data

For longitudinal studies, it is important to ensure variable names are comparable across waves. This can be a complex process so a variable naming strategy should be devised from an early stage and documented.

This should include a consistent approach to identify variables from:

- different waves of data collection

- different methods of data collection, e.g. face-to-face interview, self-completion, clinical tests
- different respondents, e.g. cohort member, parent, teacher, nurse.

Comparable variables should have consistent names across waves, i.e. administrative variables such IDs, survey weights, responses from the same survey question.

It is good practice to document the variable naming strategy. This will encourage the naming strategy to be consistent over time. It is helpful to include information in the user guide, as this will help users understand how the variables are named.

3.1.2 Variable names - Understanding Society example

Understanding Society has a distinct naming convention for its variables to ensure users can identify which wave the data is from and the source of the data. This follows the naming convention used for its data files too. Some of the rules they use are:

Naming convention	Examples
Begin with a prefix to reflect the wave the data are collected	"**a_**" for the first wave; "**b_**" for the second wave; "**c_**" for third wave etc.
Use the same root name which is fixed over time	**jbstat** for current employment status; **racel** is used for ethnic group; **marstat** is used for legal marital status.
Follow wave prefix with additional prefix to identify different groups	"**sc**" for self-completion interview; "**ya**" for interview with young adults; "**cd**" for child development module.

This strategy is clearly documented in the Understanding Society user guides and on their websites. This helps new users understand the data and existing users can easily locate the variables of interest when the latest wave of data is released.

3.2 Variable labels

Variable labels should provide information about the contents of the variable. They can be embedded in the data if your software permits. Separate codebooks are also useful as they can provide extra information and enable transfer between file formats.

Variable labels should:

- Indicate the unit of measurement, where applicable (e.g. height in cm, or time in minutes).
- Reference the question number which can be cross-referenced with the questionnaire, where applicable.
- Be brief with a maximum of 80 characters. Stata will truncate variable labels longer than 80 characters.
- Exclude non-ASCII characters that extend beyond the basic English alphabet and include symbols, accented letters, and characters from different languages.

The example variable labels below show the good practice tips we have just discussed. Including the units of measurement and referencing the question number automatically increases the usability of the variable. The brief labels, including standard characters, will be easily read by users and different software.

Variable name	Variable label	Good practice to include:
q17httw	Q11: Hours spent travelling to work in a typical week	Unit of measurement and a reference to the question number.
sc20mmj	Occupation in main job, SOC2020 major group	Standard classification coding, including the date and hierarchical level (i.e. major).
heightcm	Height without shoes: cms	Unit of measurement and a reference to useful methodology (i.e. excludes shoes)
a_marstat	Marital status of cohort member at wave 1	Who is interviewed and reference to the wave of data collection for longitudinal studies.

3.3 Value labels

Value labels should provide information about the meaning of values in categorical variables. For example, a simple marital status variable might have responses with values of 1, 2, 3 and 4. Anyone looking at the data would not know what each value meant. They could guess but

without value labels embedded in the data, or the information provided in the codebook or questionnaire, a user would not be certain of the meaning of the data.

It is worth mentioning that defining value labels within your software, will not change the recorded values, but instead allow the value label description to be shown alongside the original codes. This can be useful for certain analyses and particularly helpful to new users of the data.

The best practice for value labels is very similar to variable labels. We suggest value labels should:

- Be brief with a maximum of 120 characters. SPSS will truncate variable labels longer than 120 characters.
- Exclude non-ASCII characters that extend beyond the basic English alphabet and include symbols, accented letters, and characters from different languages.

3.4 Standard classifications and coding schemes

It is good practice to use standard classifications and coding schemes within your data. They might be used during data collection and form part of the questionnaire, or they could be used during the creation of derived variables. Standard classifications and coding schemes provide a way of presenting information in a meaningful, systematic and standard format. This can aid comparability which is especially important for longitudinal data.

From a data documentation perspective, it is important to document:

- The classification scheme used. Schemes exist in the UK and internationally, so it is important to reference the classification used. You should reference the classification in the variable label and/or codebook.
- The version of the classification scheme used. Standard classifications are regularly reviewed to ensure they are still relevant. The version used should be included in the variable label and/or codebook when referencing the classification scheme used.
- Any variation from the standards. Sometimes data does not fit exactly into standard classifications and coding schemes. If there is any variation from the standard in your data, this should be noted in the data documentation, such as the technical report or user guide.

3.4.1 UK classifications

The Office for National Statistics sets out a number of classifications and standards used in UK statistics.

The Government Analysis Function department provides harmonised standards and guidance on a wide range of topic groups

3.4.2 International classifications

A number of International standard classifications exist which can be used for cross-country comparisons:

- International Standard Classification of Occupations (ISCO)
- International Standard Classification of Education (ISCED)
- International Statistical Classification of Diseases and Related Health Problems (ICD)
- International Classification of Functioning, Disability and Health (ICF)
- International Classification of Health Interventions (ICHI)
- Nomenclature of territorial units for statistics (NUTS)
- The United Nations Statistics Division is the custodian of a number of key international classifications.
- The World Health Organisation provides further derived classifications are extensions of the core reference classifications that allow for further depth within a particular area or setting.

3.5 Missing values

Missing data occur when data are not available for certain variables or participants. There are numerous reasons why a response may not have a valid code or value. It is important to follow a standardised approach and document how missing data have been treated.

3.5.1 System missing

Statistical programs like SPSS, Stata and R have specific ways of tagging missing data. You need to consider whether your data includes different categories of missing values and if they are important for your analysis. If you do not want to include the missing data, you can convert your missing values to **system missing**, so these values are excluded from the analysis. System missing in Stata and SPSS are denoted with a full stop "." and in R with "NA".

3.5.2 User defined missing data

User defined missing data occur when certain values of a variable are allocated to mean 'missing' by the user. This allows the data to show the reason behind the different types of missing values. For example, it is common in longitudinal population studies to have categories of missing data assigned to minus numbers (e.g., -9 = No response; -8 = Refusal) or large numbers (e.g., 99 = Not applicable). Consideration should be given to whether these values are treated as missing by your statistical program and if you want to include these values in your analysis. Each statistical program has a different way of dealing with missing values, so a common approach needs to be applied to all data sources. The missing value labels should be included embedded in the data file, if appropriate to your software, or provided in the codebook or user guide.

3.5.3 Understanding Society missing values example

An example of missing values used by the longitudinal survey Understanding Society, highlights the many different reasons why survey responses may not have a valid code or value. They use negative numbers which are never used as valid responses in analysis.

Understanding Society
THE UK HOUSEHOLD LONGITUDINAL STUDY

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  - Connecting family members and households over time
  - International comparisons
  - Revisions to previous releases
- Using the BHPS
- Questionnaires
- Derived Income variables
- Weighting guidance
- Analysis advice for mixed mode data
- Support and resources

Missing values

For various reasons survey responses may not have a valid code or value.

The missing value codes asigned to these data are described below. All missing values are negative and are never used as valid responses. We recommend that users carefully read the questionnaires and compare missing value distributions across waves before using the substantive information contained in them.

Value	Description
-1	"Don't know" – When the respondent does not know the answer.
-2	"Refused" – When the respondent refuses to answer a question.
-7	"Proxy" – Sometimes when a person cannot participate in the interview, someone else in the household (generally their spouse or partner or adult children) answers questions on their behalf, that is, by proxy. This questionnaire is a much shorter questionnaire asking factual information. So, if a question was not included in the proxy questionnaire and the person gave a proxy interview, this variable will be missing for them. In such cases the variable will have a value of -7.
-8	"Valid skip" – This information is missing because the person was never asked this question as they were not eligible for it. E.g., someone who is not in paid employment is not asked questions about their pay.
-9	"Missing by error or implausible".
-10	"Not available for the IEMBS" – Some questions were only asked in the W6 questionnaire for non-IEMBS and so individuals in IEMBS will have missing information for these variables.
-11	"Only available for the IEMBS" – Some questions were only asked in the W6 questionnaire for IEMBS and so individuals in non-IEMB samples will have missing information for these variables.
-20	"No data from the BHPS W1-18" – This code is only used for variables in the xwavedat file which is harmonised across BHPS and UKHLS. If some variable was only asked in the UKHLS then there will be no data from BHPS W1-18 and hence missing for those not interviewed during UKHLS.
-21	"No data from the UKHLS". – This code is only used for variables in the xwavedat file which is harmonised across BHPS and UKHLS. If a variable was only asked in the BHPS then there will be no data from UKHLS and hence missing for those not interviewed during BHPS W1-18.

3.6 Derived variables

Derived variables very much enhance the value of a dataset and are much appreciated by researchers. They provide new information as well as save time and effort.

Derived variables are created by calculating or categorising variables that already exist in your data. They can involve a mathematical calculation (e.g. deriving Body Mass Index from height and weight variables) or a re-categorisation of one or more existing variables (e.g. categorising single year of age into grouped bands - Under 16, 16-19, 20-24, 25-29, etc. or

creating a household composition variable using a mixture of sex, marital status and age variables). Derived variables can be particularly useful for longitudinal research as they can provide cross-wave information in an easy-to-use format.

Derived variables can range from simple recodes to extremely complex calculations and re-categorisations. It is vitally important to provide documentation about derived variables to help users understand how the variables have been created and ensure replicability in future waves.

It is good practice to document the following information:

- **Provide details of derivation.** This should ideally be the syntax used to create the derived variable. If is not suitable to share the syntax, you should include information about the source variable(s) and any standard classifications and coding schemes used. This information can be provided in the user guide, or, if you have a large number of derived variables, in a separate derived variable user guide.
- **Flag the variable is derived in the variable name.** It is helpful for users to identify easily which variables have been derived. A common approach is to include 'dv' in the variable name, but this might not be appropriate for larger cross-wave surveys, so a more structured approach could be used and documented.
- **Consider separating derived variables in the file structure.** Derived variables are often placed at the end of the data file. This helps users to identify these variables. For larger studies, a separate data file containing all the derived variables could be useful. The size and structure of your data will dictate the best approach.

Activity: Data without Context (Qualitative data)

Please have a look at the screenshot of this interview transcript extract taken from the UKDS QualiBank. What information is missing?

Exercise – Data without context (qual.)

I: If I could ask you when you were born?
R: Kings College Mews.
I: I see, when was it?
R: 1887.
I: And what did your father do?
R: My father was, well really a navvy. And he worked on these houses in Wadham Gardens, Harley Road, this top end. And when we were children we used to go round and pick up firewood. For fires. Yes.
I: How many were there in your family?
R: In my time, 9.
I: 9 children?
R: Yes.
I: How many rooms had you?
R: 3.
I: It must have been a bit of a squash.
R: Well it was a squash, in those days. Yes, of course the rents was different those days.
I: How much were you paying?
R: Best I recollect my mother paying was 7/6d. Yes.
I: She hadn't any help?
R: In the home, no, no.
I: She managed by herself in this house. It was actually this house was it, where you were?
R: Not in this house, we've moved from one house to another. As the family got larger, then of course mother moved about.
I: I see.
R: We were in No. 10. Then we went to 13. I think my mother stayed there when we broke away from home, marrying, you know.
I: She moved up and down on one row of houses?
R: Mother died there. And Father died there. 'Course they were a good age, I can't remember their ages.
I: Whereabouts were you in your family? Were your brothers and sisters older or younger or what?
R: Some were older than me and some, I got 1, 2, 3, younger than me.

Source: Thompson, P., University of Essex. Department of Sociology, Lummis, T., University of Essex. Department of Sociology: "Interview with Mr. Rook" in "Family Life and Work Experience Before 1918, 1870-1973" 7, UK Data Service [distributor], 2009-05-12, SN:2000, Para. 1-26. DOI: 0.5255/UKDA-SN-2000-1, discover.ukdataservice.ac.uk//QualiBank/Document/?cid=q-cdb2a4f4-ebb8-40d2-bf6a-a098bf712e17

Answers should include:
More information on the Respondent (R)
More information on the Interviewer (I)
Data listing
Background information about the study
Dates of fieldwork
Method of data collection
Sampling procedures
Population
User guide
Technical report.
(All should be forms of documentation (data-level, metadata or study/project-level).)

Exercise – Data without context (qual.)

Interview with Mr. Rook

External resources | Collection documentation Access this collection from the Data Catalogue

DETAILS

Collection ID (SN):	2000
Title:	Family Life and Work Experience Before 1918, 1870-1973
Principal investigator:	Thompson, P., University of Essex. Department of Sociology Lummis, T., University of Essex. Department of Sociology
Sex:	Male
Age group:	75-84
Socio-economic status:	Routine
Region:	London
Licence:	Creative Commons Attribution 4.0 International License

Additional metadata

Idno:	2000int008
No. pages:	82
Placename:	London, Greater London area
R title:	Mr.
R forename:	James
R surname:	Rook
R marital status:	Married
R occupational group:	Unskilled manual
R occupation:	Porter

Basic metadata which is available alongside the interview from QualiBank.

This example highlights how providing some key metadata, such as the study title, fieldwork dates and basic demographic information about the respondent, adds more meaning to the interview transcript extract.

Good documentation is essential to ensuring data can be understood and reused. Users need sufficient contextual and explanatory information to make sense of the data.

4. Qualitative Data-level documentation

Qualitative data-level documentation describes the contents of an individual data file. This could be an interview or focus group transcript, or an audio, image or video file, or a combination of different data types. This documentation can either be embedded in the data file or provided alongside the data files, in a structured data list.

4.1 Embedded metadata

Documentation can be embedded in your data file. The type of data will influence how this can be done but some common examples are:

- **Header or summary page.** For textual data, information can be added at the beginning of a file as a header or summary page. This might include background information about the respondent and information about the data collection process. This adds contextual value to the data and is useful for secondary users.
- **File properties.** Most files types such as Microsoft Office files or Adobe PDF have ‘properties’ section in the menu. Information can either be captured automatically or

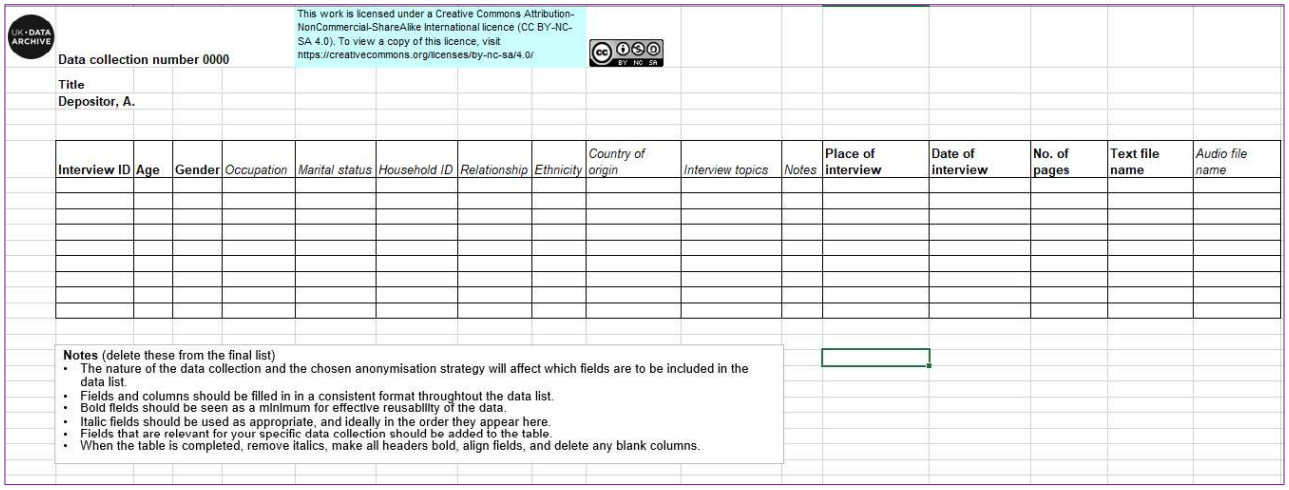
entered manually. Basic information such as file names, title and keywords can be entered or stored. Personal information should not be included due to confidentiality issues.

- **Within qualitative data analysis software.** Using a qualitative data analysis (QDA) software package, such as NVivo or Atlas.ti, can help you to organise, code and analyse your data. Users can add codes and classifications to their data files and this information can be exported in external files too.

4.2 Data list

Qualitative research is likely to generate a lot of data files. These might include a combination of data types, such as interview transcripts, audio files and images, which are linked. A data list provides an overview of the whole data collection and allows relevant items to be easily identified.

The UK Data Service website includes an data list template which can be used as a good starting point. The list provides key biographical characteristics and features of interviewees, plus details for the interview.



UK DATA ARCHIVE

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Data collection number 0000

Title

Depositor, A.

Interview ID	**Age**	**Gender**	*Occupation*	*Marital status*	*Household ID*	*Relationship*	*Ethnicity*	*Country of origin*	*Interview topics*	*Notes*	**Place of interview**	**Date of interview**	**No. of pages**	**Text file name**	*Audio file name*

Notes (delete these from the final list)
- The nature of the data collection and the chosen anonymisation strategy will affect which fields are to be included in the data list.
- Fields and columns should be filled in in a consistent format throughtout the data list.
- Bold fields should be seen as a minimum for effective reusability of the data.
- Italic fields should be used as appropriate, and ideally in the order they appear here.
- Fields that are relevant for your specific data collection should be added to the table.
- When the table is completed, remove italics, make all headers bold, align fields, and delete any blank columns.

Some key points to remember regarding data lists are:

- Each item in the list should have a unique identifier. Identifiers used should be consistent so links can be made between interview transcripts, interview recordings, field notes, etc.
- Columns can be added or removed to ensure the information included suits your data collection.
- The list should indicate where parts of the data are missing, such as partial or missing transcripts.
- No disclosive information should be included, so where appropriate, pseudonyms should be used to anonymise participants.

This example shows part of a populated data list from a collection available at the UK Data Archive. A unique identifier is included and the text transcript files and the audio files are clearly labelled. Extra columns have been added to provide information specific to this data collection such as details of the farms. Some audio files are missing but this is clear from the data list. No identifying information is included.

Study Number 6377
Integrated Floodplain Management, 2006-2008
Morris, J.

Floodplain farm survey

Interview ID	Farmer code	Age	Farm scheme	Farm type	Size of farm (hectare)	Number of holdings	Date of interview	Interviewer name	No of pages	Text file name	Audio file name
1	Be1	35-45	Beckingham	Beef	360	1	04.12.2006	Helena	28	6377int001	6377int001
2	Be2	45-55	Beckingham	Arable	364	1	05.12.2006	Helena	21	6377int002	6377int002
3	Be3	45-55	Beckingham	Arable	372	2	06.12.2006	Helena	22	6377int003	6377int003
4	Be4	45-55	Beckingham	Arable	194	3	06.12.2006	Helena	18	6377int004	6377int004
5	Be5	55-65	Beckingham	Arable	108	1	07.12.2007	Helena	21	6377int005	6377int005
6	Be6	45-55	Beckingham	Arable	1254	2	01.02.2008	Helena	19	6377int006	
7	Bu1	55-65	Bushley	Mixed	101	2	13.02.2007	Quentin	29	6377int007	6377int007
8	Bu2	>65	Bushley	Mixed	97	1	15.02.2007	Quentin	15	6377int008	6377int008
9	Bu3	>65	Bushley	Arable	194	4	13.02.2007	Quentin	21	6377int009	6377int009
10	Bu4	55-65	Bushley	Mixed	202	1	15.03.2007	Helena	19	6377int010	6377int010
11	Cu1	35-45	Cuddyarch	Dairy	64	1	08.05.2007	Helena	19	6377int011	6377int011
12	Cu2	55-65	Cuddyarch	Dairy	189	2	08.05.2007	Helena	18	6377int012	6377int012
13	Cu3	55-65	Cuddyarch	Mixed livestock	76	1	08.05.2007	Helena	13	6377int013	6377int013
14	Cu5	45-55	Cuddyarch	Mixed livestock	198	1	09.05.2007	Helena	24	6377int014	6377int014
15	Cu6	55-65	Cuddyarch	Dairy	89	1	09.05.2007	Helena	14	6377int015	6377int015
16	Cu7	>65	Cuddyarch	Mixed livestock	190	4	11.05.2007	Helena	20	6377int016	6377int016
17	Cu8	55-65	Cuddyarch	Mixed livestock	109	2	11.05.2007	Helena	22	6377int017	6377int017
18	Id1	55-65	Idle	Arable	158	3	07.02.2007	Quentin	17	6377int018	6377int018a
18	Id1	55-65	Idle	Arable	158	3	07.02.2007	Quentin	17	6377int018	6377int018b
19	Id1b	55-65	Idle	Arable	158	3		Quentin	22	6377int019	
20	Id2	45-55	Idle	Dairy	150	1	08.02.2007	Quentin	17	6377int020	6377int020

5. Study-level documentation

Study-level documentation for a data collection or dataset provides high-level information on the research context and design, the data collection methods used, any data preparations and manipulations, plus summaries of findings based on the data.

5.1 Best practice

Good study-level data documentation should include information on the following:

- **Purpose for which data was created:** Project history, aims, objectives and hypotheses, investigators, funders and publications from the data.
- **Content of the dataset:** Kind of data (e.g. interviews, images, survey responses), the structure of data files, with the number of cases, records, files and variables, as well as any relationships among such items.
- **How the data was collected**: Data collection methodology and protocols, sampling design, sample structure and representation, workflows, instruments used, hardware and software used, data scale and resolution, digitisation or transcription methods used, secondary data sources used.

- **Who collected the data, where and when:** Data collectors, geographic coverage of the data collection, date of data collection (temporal coverage).
- **How data was processed:** Tools, instruments, procedures, hardware/software or protocols used to edit, clean, code and classify the data.
- **Any modifications done to the data:** Anonymisation, for time series or longitudinal surveys changes made to methodology, variable content, question text, variable labelling, measurements or sampling.
- **Quality assurance procedures:** Data validation, checking, proofing, cleaning and other quality assurance procedures carried out, such as checking for equipment and transcription errors, calibration procedures, data capture resolution and repetitions, or editing, proofing or quality control of materials.
- **How can data be accessed:** Where is the dataset available from, persistent identifier, access conditions, use conditions, licence, copyright, information on data confidentiality, citation.

In reality, a number of guidance documents will need to be created. The size of your study and data collection processes involved will influence the amount of suitable study level documentation, but we suggest each data collection should include the following as a minimum:

- User guide or Technical report.
- Questionnaire.
- Codebook.

Longitudinal studies should also consider including:

- **Sweep/wave-level documentation:** This should contain similar information to study-level documentation, but specific to a certain wave of data collection.
- **Linkage documentation:** If data from other sources have been linked to a dataset, the linkage documentation describes the linkage procedures, including linkage variables, matching algorithms, and privacy protection measures. This can also cover information on linking sweeps/waves.

5.2 UK Data Service study-level documentation - Example

Home > Data catalogue > Studies > Study

Young Persons' Behaviour and Attitudes Survey, 2022

Details | Documentation | Resources | Access data

Documentation

Title	File name	Size (MB)
UK Data Archive Citation File for Study 9195	UKDA_Study_9195_Information.htm	0
UK Data Archive Data Dictionaries	ukda_data_dictionaries.zip	0.16
UK Data Archive ReadMe File for Study 9195	read9195.htm	0
YPBAS 2022 Database Codebook	9195_ypbas_2022_codebook.pdf	0.81
YPBAS 2022 Key Findings	9195_ypbas_2022_key_findings.pdf	0.65
YPBAS 2022 Questionnaire Version A	9195_ypbas_2022_questionnaire_version_a.pdf	0.92
YPBAS 2022 Questionnaire Version B	9195_ypbas_2022_questionnaire_version_b.pdf	1.03
YPBAS 2022 Tables	9195_ypbas_2022_tables.xlsx	0.11
YPBAS 2022 Technical Report	9195_ypbas_2022_technical_report.pdf	0.53
YPBAS Weights on 2022 Microdata	9195_ypbas_2022_weights_on_microdata.pdf	0.24

This example shows the sort of study-level documentation provided for a typical study available from the UKDS. This includes a Technical Report, Questionnaires, a Codebook, a Key Findings report and a separate guide about the weighting used.

The UKDS also provides users with a number of files as part of our standard curation process:

- A **Citation file** to encourage correct citation, including a Digital Object Identifier (DOI).
- A **ReadMe file**, containing details of our curation processes, specific information noted during our checks, and information of the conditions of use.
- **Text Data Dictionaries** generated during our curation processes which include file level information such as file name and number of variables and cases, and variable level information such as position in data file, name, labels, type, measurement level and details of any missing values.

6. Metadata

We are now going to look at metadata in detail. Metadata helps users find, understand, and use data, but it is important to understand and define metadata.

Metadata is often referred to data about data. However, this definition is not particularly useful if you are not already clear what 'data' are and how to interpret them.

Instead, it is more useful to think of the difference between data and information: for data to become information, you need to understand the context in which the data are situated. Metadata is what provides this essential context.

Metadata + Data = Information

Exercise: If we look at this data table, can you understand what the data represent?

1	18	175	Brown	71
2	20	163	Green	53
2	17	165	Green	57
1	23	180	Blue	78
1	21	184	Blue	82

We might be able to guess the meaning, but we need extra information to help us to understand and use the data.

By adding additional metadata, the data table is suddenly transformed, and the usability has dramatically increased.

Sex	Age	Height	Eye colour	Weight
1	18	175	Brown	71
2	20	163	Green	53
2	17	165	Green	57
1	23	180	Blue	78
1	21	184	Blue	82

However we are still missing some important pieces of contextual information. What is missing?

Sex	Age	Height	Eye colour	Weight
1=Male, 2=Female	**(years)**	**(centimetres)**		**(kilograms)**
1	18	175	Brown	71
2	20	163	Green	53
2	17	165	Green	57
1	23	180	Blue	78
1	21	184	Blue	82

Applying additional metadata provides further information about the units in which numerical data are presented and makes the data even easier to interpret.

6.1 Creating metadata during the research data lifecycle

The example we have just looked at highlights the importance of metadata for the data file, but metadata should cover the whole research data cycle. To really use and understand the data in our example table, you would need to know additional information.

If we look at the '5Ws' and '1H', which are commonly used in journalism to gather information, we will begin to see the sort of information we need to know in order to use and understand the data correctly.

- **Who created the data?** (Citation and information on data curators, data collectors, sponsors and contributors)
- **What does the data file contain?** (Data format, file type, file structure, variable descriptions, missing data, weighting variables, software and keywords)
- **When were the data created?** (Dates of fieldwork, time period covered, time dimension (e.g. cross-sectional or longitudinal)
- **Where were the data created?** (Spatial units and geographical coverage)
- **Why were the data created?** (Background to creation and planned use)
- **How were the data created?** (Methodology of data collection, sampling and processing, data access information)

Metadata provides the answers to these questions but in a structured format which can be read by machines.

We are going to look at structured metadata in more detail.

6.2 Structured metadata

Structured metadata defines the relationship between data items to enable computer systems to understand the contextual meaning of the data - to display the relevant information on a website, for instance.

Structured metadata tells a computer what something is, how it relates to other objects and what to do with it. By **standardising** the **content** and **structure**, it makes it easier for computers to automatically extract **information** from the metadata.

This information can then be provided to researchers to help them discover and access data from many different sources. It facilitates data sharing and allows data collected in one study to be re-used in the future by other researchers.

6.3 Metadata standards

A metadata standard provides a framework to establish a common set of definitions for various characteristics or attributes of data. Standardising metadata, including language, spelling, format, variable coding, etc., allows different datasets to 'speak' to each other. If everyone uses a different standard, it can be very difficult to compare data from different sources. Because there is not one overall standard, it is necessary for 'translation' programmes to map between standards. In this way systems can read datasets using different standards.

Metadata standards allow:

- re-use of data
- data discovery
- data access
- interoperability of systems - that is, systems and machines can talk to each other and know they are referring to the same thing
- sharing of metadata between communities (e.g. data providers and data users).

Certain metadata standards are necessary to harvest metadata into other portals and thus play an important role for discoverability, nationally as well as internationally.

6.4 Example metadata schemas

Ideally, metadata is produced and presented using standardised specifications. This again facilitates the re-use of data, data discovery, data access and the ability to share metadata between communities of researchers.

- Dublin Core: The first metadata standard for describing web content.
- ISO 19115 -1:2014: For describing geographic data and services.
- Data Documentation Initiative (DDI): For describing the data produced by surveys and other observational methods in the social, behavioural, economic, and health sciences.
- Statistical Data and Metadata eXchange (SDMX): An ISO standard designed to describe statistical data and metadata, normalise their exchange, and improve their efficient sharing across statistical and similar organisations.
- Metadata Encoding and Transmission Standard (METS): For encoding descriptive, administrative, and structural metadata regarding objects within a digital library, expressed using the XML schema language of the World Wide Web Consortium (W3C)

- General International Standard Archival Description (ISAD(G)): An international standard which provides guidelines for creating the content of an archival description.
- DataCite metadata schema: For the publication and citation of digital datasets with a persistent identifier.

6.5 Data Documentation Initiative (DDI)

The DDI is a rich and detailed metadata standard, developed by the international DDI Alliance. It was originally designed for describing social, behavioural and economic sciences data. It is used by many social science organisations across the world.

There are several version of DDI. Firstly, **DDI Codebook (DDI-C)** gathers basic information aimed at describing data and is generally used to compile catalogue records. **DDI-Lifecycle (DDI-L)** is a more complex schema, and contains additional elements used to describe survey questions and variables, as well as basic catalogue information. This is particularly useful for longitudinal studies, where the same questions and variables may be used across survey waves. Some longitudinal studies use software such as Colectica to manage their documentation and data. This gathers formatted metadata in DDI-L format, enabling searching across waves and between surveys. For example, the CLOSER Discovery metadata system has been developed using DDI-L.

DDI catalogue records contain mandatory and optional metadata elements relating to study description, data file description and variable description. The study description contains information about the context of the data collection, such as bibliographic citation of the study and data, the scope of the study, like topics, geography, time, method of data collection, sampling and processing, data access information, and information on accompanying materials.

The data file description indicates data format, file type, file structure, missing data, weighting variables and software used. Variable descriptions indicate the variable labels and codes.

For example, the UK Data Service uses DDI to structure catalogue records and standardised records in eXtensible Mark-up Language (XML). This brings key data documentation together into a single document, creating rich and structured content about the data, and metadata can be harvested for data sharing through the Open Archives Initiative Protocol for Metadata Harvesting (OAI-PMH).

6.6 DDI and archive catalogue records

If you are planning to archive data (or set up their own catalogue), you will need to provide metadata. At the UK Data Service, our DDI records contain mandatory and optional metadata elements on the following areas:

- **Study description** - information about the context of the data collection, such as bibliographic citation of the study and data, the scope of the study (topics, geography,

time), methodology of data collection, sampling and processing, data access information, and information on accompanying materials.

- **Data file description** - information on data format, file type, file structure, missing data, weighting variables and software.
- **Variable descriptions**.
- **Keywords** - it is good practice to use a thesaurus such as ELSST (European Language Social Science Thesaurus), the multilingual thesaurus for the social sciences.

A useful tool is the CESSDA Metadata Model (CMM) which gives guidance on typical metadata elements. The tool contains metadata elements, their definitions and information on other requirements, such as repeatability. The Model is built from the viewpoint of quantitative (social science) data and is based on DDI Lifecycle 3.2 metadata standard.

Whether you use any of the metadata standards we've described here or not, it's worth considering how to describe and share your metadata as openly as possible. Using the FAIR principles can help you with this (we'll talk more about this below), and other tools, such as metadata profiles, are available to help you decide on the important metadata elements to include when describing your study.

6.7 Metadata profiles

Alongside the metadata model, CESSDA have made available a range of metadata profiles for various types of DDI. While they are primarily intended for CESSDA member organisations who contribute records from their own catalogues to the CESSDA Data Catalogue, the profiles can be used by anyone to see the elements of metadata that need to be included to construct a comprehensive description of your data - title, author, dates, coverage, population and sampling, methods used to collect the data, etc.

The DDI Alliance also provides information on best practice for creating a metadata profile.

7. Resources

UKDS Learning Hub Research data management
ukdataservice.ac.uk/learning-hub/research-data-management/#document-your-data

CESSDA Data Management Expert Guide (DMEG)
dmeg.cessda.eu/Data-Management-Expert-Guide/2.-Organise-Document/Documentation-and-metadata

Managing and sharing data: best practice for researchers
dam.ukdataservice.ac.uk/media/622417/managingsharing.pdf

Kaye, N., Mills, H. & Johnson, J. (2020). Understanding metadata. CLOSER Learning Hub, London, UK: CLOSER
learning.closer.ac.uk/learning-modules/understanding-metadata/

GO FAIR - FAIR Principles
www.go-fair.org/fair-principles/

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This project is funded by the UKRI Economic and Social Research Council and the UKRI Medical Research Council.